CCE-(Unit-5)

1. **Problem Statement**

   Develop a C++ program to represent hospital equipment units as vertices and possible wiring links as weighted edges. Apply Kruskal's algorithm to select the minimum-cost wiring connections. Display selected edges and total wiring cost, and list edges rejected because they create cycles.

2. **Solution**

   The hospital equipment wiring problem is solved using Kruskal's Algorithm. Each hospital equipment unit is represented as a vertex, and each possible wiring connection is represented as a weighted edge.

   First, all edges are sorted in ascending order of their costs. The algorithm selects the lowest-cost edge that does not create a cycle. The Disjoint Set Union (DSU) data structure is used to detect cycles. The process continues until all equipment units are connected or all edges have been checked.

   Finally, the program displays the selected wiring connections, the minimum total wiring cost, and the rejected edges that create cycles. This approach helps connect hospital equipment with minimum cost while avoiding unnecessary wiring connections.

3. **. Objectives**

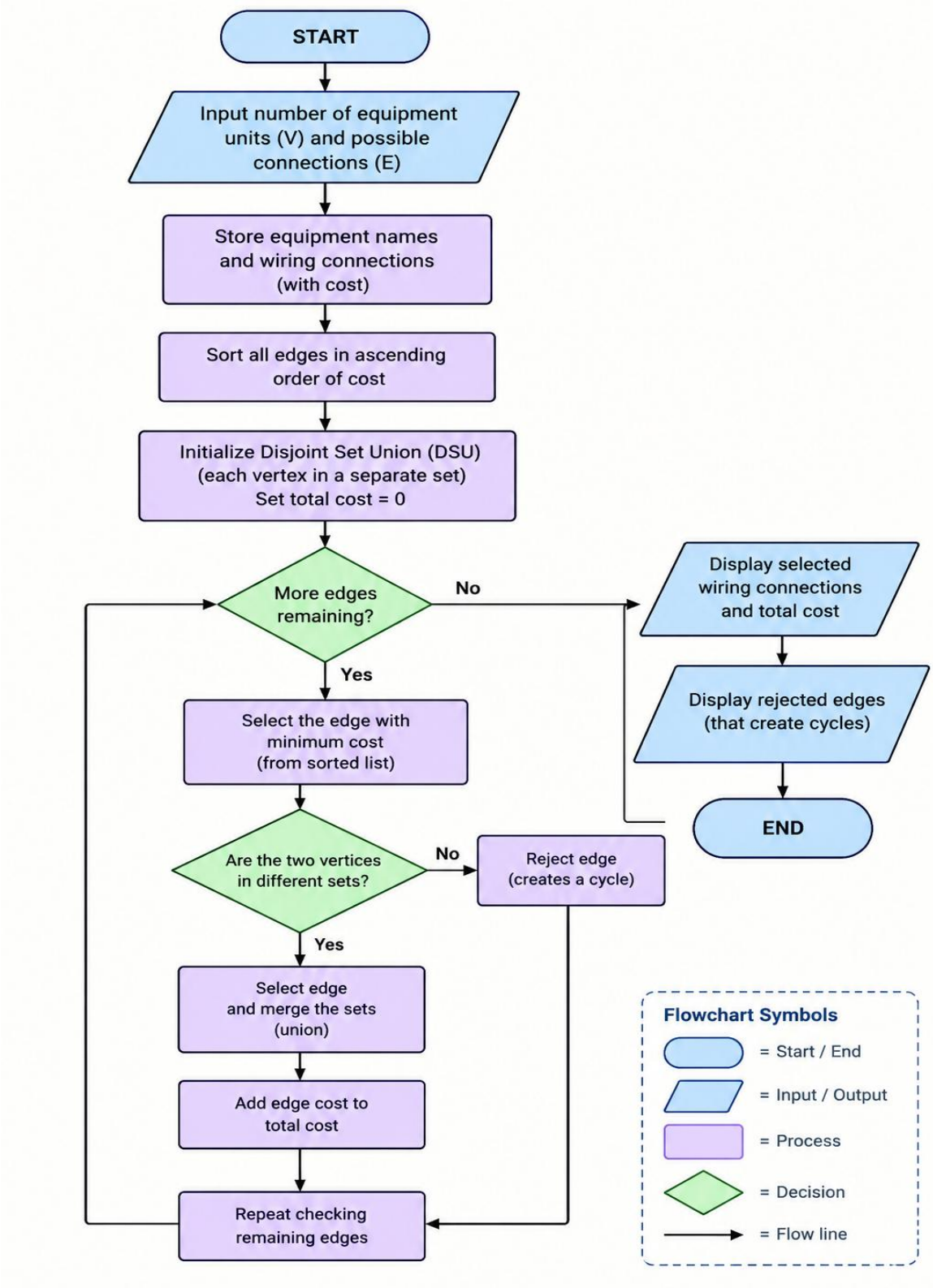
1. To represent hospital equipment units as vertices in a graph.
2. To represent possible wiring connections as weighted edges.
3. To implement Kruskal's Algorithm using C++.
4. To find the minimum-cost wiring connections.
5. To identify edges rejected because they create cycles.
6. To calculate and display the total wiring cost.

4. **Algorithm**

1. Start the program.
2. Input the number of hospital equipment units and possible wiring connections.
3. Assign a name or number to each equipment unit.
4. Store each wiring connection with its source, destination, and cost.
5. Sort all edges in ascending order of cost.

6. Initialize each vertex as a separate set.
7. Select the edge with the minimum cost.
8. Check whether its endpoints belong to different sets:
    - If yes, select the edge and merge the two sets.
    - If no, reject the edge because it creates a cycle.
9. Repeat until V - 1 edges have been selected or all edges have been processed.
10. Display the selected edges and total wiring cost.
11. Display the rejected edges that create cycles.
12. Stop.

5. Flowchart


START
Input number of equipment units (V) and possible connections (E)
Store equipment names and wiring connections (with cost)
Sort all edges in ascending order of cost
Initialize Disjoint Set Union (DSU) (each vertex in a separate set) Set total cost = 0
More edges remaining?
No
Yes
Display selected wiring connections and total cost
Display rejected edges (that create cycles)
END
Select the edge with minimum cost (from sorted list)
Are the two vertices in different sets?
No
Reject edge (creates a cycle)
Yes
Select edge and merge the sets (union)
Add edge cost to total cost
Repeat checking remaining edges
Flowchart Symbols
= Start / End
= Input / Output
= Process
= Decision
= Flow line